

## Question

In the crystal of metal fluoride  $\text{MF}_n$ ,  $\text{MF}_4$  quadrilaterals are vertex-sharing connected to form a helical structure, where each helix is independent, and a  $6_1$  screw axis passes through the unit cell point  $(0, 0, 0)$ .

The coordinates of one  $\text{M}$  are known to be  $(0.4812, 0.2406, 0.0833)$ , and the coordinates of an F atom shared by two  $\text{M}$  atoms are  $(0.1673, 0.1673, 0.1667)$ .

Around the  $\text{M}$  atom at coordinates  $(0.4812, 0.2406, 0.0833)$  (denoted as  $\text{M}_1$ ), there are two distinct  $\text{M} - \text{F}$  bond lengths (with a ratio of 1:1, measuring 205.2 pm and 193.3 pm, respectively).

Which of the following statements is correct?

**A.** The bond length between the F atom at coordinates  $(0.1673, 0.1673, 0.1667)$  and  $\text{M}_1$  is 193.3 pm.

**B.** Crystal coordinate parameters:  $a = b = 534.5$  pm,  $c = 1652.6$  pm

**C.** There is an  $\text{M}$  atom at  $(0, 0.1673, 0.3333)$ .

**D.** There is an  $\text{M}$  atom at  $(0.7594, 0.2496, 0.4167)$

**E.** If the density of the crystal is  $6.189 \text{ g/cm}^3$ , then  $\text{M}$  is most likely *Re*.

**F.** None of the above statements are correct.

**G.** The above statements are multiple correct

## Answer

Correct Answer: **B**

## Detailed Explanation

According to the question, the chemical formula of the crystal is  $\text{MF}_3$ , which belongs to the hexagonal crystal system.

### CHECKPOINT

1 PTS

The chemical formula of the crystal is  $\text{MF}_3$

According to the information in the question, some coordinates of  $M$  and  $F$  can be deduced:

$M(0.4812, 0.2406, 0.0833)(0.2406, 0.7594, 0.9167)(0.7594, 0.5188, 0.7500)(0.5188, 0.7594, 0.5833)(0.7594, 0.2406, 0.4167)(0.2406, 0.4812, 0.2500)$

### CHECKPOINT

1 PTS

$M(0.4812, 0.2406, 0.0833)(0.2406, 0.7594, 0.9167)(0.7594, 0.5188, 0.7500)(0.5188, 0.7594, 0.5833)(0.7594, 0.2406, 0.4167)(0.2406, 0.4812, 0.2500)$

$F(0.1673, 0.1673, 0.1667)(0, 0.1673, 0.3333)(0.8327, 0, 0.5000)(0.8327, 0.8327, 0.6667)(0, 0.8327, 0.8333)(0.1673, 0, 0)$

### CHECKPOINT

1 PTS

$F(0.1673, 0.1673, 0.1667)(0, 0.1673, 0.3333)(0.8327, 0, 0.5000)(0.8327, 0.8327, 0.6667)(0, 0.8327, 0.8333)(0.1673, 0, 0)$

Therefore,  $CD$  is incorrect.

The  $F$  atom with coordinates  $(0.1673, 0.1673, 0.1667)$  is a bridging atom, and its bond length corresponds to 205.2pm

### CHECKPOINT

1 PTS

The  $F$  atom with coordinates  $(0.1673, 0.1673, 0.1667)$  is a bridging atom, and its bond length corresponds to 205.2pm

Therefore,  $A$  is incorrect.

$$205.2\text{pm} = [(0.0834c)^2 + (0.3139a)^2 + (0.0733a)^2 - 0.3139a \times 0.0733a]^{0.5}$$

$$193.3\text{pm} = [(0.0791c)^2 + (0.2949a)^2 + (0.2233a)^2 - 0.2949a \times 0.2233a]^{0.5}$$

Solving for  $a = 534.5\text{pm}$ ,  $c = 1652.6\text{pm}$

CHECKPOINT

1 PTS

$a = 534.5\text{pm}$ ,  $c = 1652.6\text{pm}$

Therefore,  $B$  is correct.

According to the density,  $M$  should be  $Au$ ,  $E$  is incorrect.

CHECKPOINT

1 PTS

$M$  should be  $Au$

In summary, choose  $B$ .